# The Rise of Online Gambling: What's at Stake?

Team #18486

## Executive Summary

Online sports betting has seen a significant increase since the Supreme Court repealed the ban on it in 2018. By 2025, the sports betting websites in the US will have generated a revenue of $15 billion from over 58 million active bettors, which accounts for 22% of the adult population. In order to determine whether the expansion of sports betting has the potential to harm society, we have developed three different models.

Q1 - How much money is really available to be spent? To understand this, the amount of disposable income available to people is measured, which is the amount of money left after taxes and essentials have been paid. This is done by studying government spending surveys from the U.S. and the UK. For someone who earns $110,000 and is halfway through their career, the amount of disposable income available to them is $13,431. For someone who earns $35,000 and is young, the amount of disposable income available to them would be negative, as their essentials would cost more than they earn. In the UK, people keep more of their earnings due to the fact that their essentials, like housing, cost less, and they also have their healthcare taken care of.

Q2 – What do gamblers stand to lose? To estimate the yearly losses for gamblers, we use two parameters: disposable income and risk tolerance level, which is on a scale of 1 to 5. A conservative gambler making simple bets can stand to lose around $60 annually. An aggressive gambler making parlay bets can stand to lose around $2,901, which is approximately 22% of the total disposable income. When aggregated across the total population in the U.S., our model corroborates the industry's reported revenue of around $15 billion with a margin of 3%.

Q3 - We created a 30-year model of how people save and accounted for the ups and downs of stocks and the random effect of gambling each year. If you do not gamble at all, your median savings will grow to $217,000. But if you are a high-risk gambler, your median savings will end up in the negatives at -$15,000, effectively in debt, 74% of the time.

In addition, our study found that high-risk gambling costs more than all other forms of entertainment combined and that 84% of high-risk gamblers lose more than 10% of their disposable income each year.

Bottom line: Gamblers, on average, lose $266 a year. But the reality behind these numbers is a disturbing trend of a few high-risk gamblers, mostly young males under the age of 35, contributing to the industry's revenue, but ruining their own futures in the process.

# 1 Q1: Playing With House Money

## 1.1 Problem

With the given salary, age, and place of residence (region and country, either the U.S. or the U.K.), the amount left after taxes and other living expenses can be estimated. The amount left after all these expenses is called disposable income.

$$\text{Disposable Income} = \text{Salary} - \text{Taxes} - \text{Essential Spending}$$

This number determines how much someone can afford to spend on things that aren't essential, such as entertainment, savings, or, in this case, gambling.

## 1.2 Assumptions

1. Single income households. We assume that the entire salary is earned by the individual, without any spousal income, investments, or government benefits coming into the picture. This is an ideal scenario, but it gives us an idea of the purchasing power that an individual's salary can generate.
2. Five essential expense categories: The essential expenses are made up of expenses that every household has to pay, which are food, housing (including rent or mortgage and utilities), transport (car loan or fuel or public transport), healthcare (premiums and out-of-pocket expenses), and insurance/pensions (life insurance and pension contributions). These comprise over 80% of the average household's non-tax expenses.
3. Government survey data. All baseline spending amounts are obtained from official government sources: the United States Bureau of Labor Statistics' Consumer Expenditure Survey for 2024 and the United Kingdom's Office for National Statistics Family Spending Survey for 2024. These surveys contain the most comprehensive official records of actual household expenditures, and we assume that they are representative of the entire population.
4. State tax approximation. US state income taxes vary greatly, ranging from 0% in Texas to 13.3% in California. However, we approximate this with a flat 5% rate, which is similar to the weighted average across the US. This is not exact for extreme states, but it keeps things simple and easy to manage.
5. Currency conversion. All U.K. currency is converted to U.S. dollars at \$1.27 = £1, so results can be compared directly.

## 1.3 How the Model Works

Our model involves two steps: calculate taxes, then estimate essential spending. From there we can calculate the disposable income by subtracting taxes and essential spending from income.

### 1.3.1 Taxes

United States: We have three types of taxes to compute: (1) FICA Payroll tax of 7.65% of salary to fund Social Security and Medicare, (2) income tax using the 2024 income tax brackets after deducting the standard deduction of \$14,600, and (3) state income tax of 5%.

The way the tax brackets work is like this: the first \$11,600 is taxed at 10%, the next amount up to \$47,150 is taxed at 12%, the amount up to \$100,525 is taxed at 22%, and so on. This means that the more you make, the more taxes you pay, but not on the whole amount.

In the United Kingdom, we see two main types of tax. The first is income tax, with a personal allowance of £12,570 tax-free, 20% tax on earnings between £12,570 and £50,270, and 40% tax on earnings above £50,270. The second is National Insurance, with an 8% rate for middle earnings and 2% for higher earnings. The result is that the United Kingdom is more beneficial for middle incomes because of the personal allowance, which is more relative to the median salary, and the lack of an equivalent tax rate like the 7.65% FICA tax.

### 1.3.2 Essential Spending

However, the basic challenge is that essential costs fluctuate wildly depending on individual circumstances. For example, a 22-year-old renter in San Francisco faces vastly different housing costs from a 60-year-old with a paid-off mortgage in Tennessee. Our model suggests that there are three main factors that contribute to these differences:

For each essential category, we estimate annual spending as:

$$\boxed{\text{Spending}_i = \text{Baseline}_i \times \text{Income Adjustment}_i \times \text{Regional Adjustment}_i} \tag{1}$$

The baseline is just the average spending for someone within this age group, and it is taken directly from government survey data (Table 1). For instance, the average American aged 25–34 spends \$26,380/year on housing as seen in the table.

The income adjustment takes into account the fact that as people earn more, they tend to spend more on the basics, although not in proportion to their earnings. A 10% increase in earnings does not mean that the amount spent on groceries will increase by 10%. This rate at which spending increases in proportion to earnings is referred to as the income elasticity of spending. This can be written as:

$$\text{Income Adjustment}_i = \left(\frac{\text{Your Salary}}{\text{Average Salary for Your Age}}\right)^{\alpha_i}$$

where $\alpha_i$ is the elasticity for category $i$. Because all elasticities are below 1.0 (see Table 3), this adjustment is always smaller than the raw income ratio. For instance, someone earning 20% more than average with a food elasticity of 0.33 spends only about 6% more on food.

The regional adjustment is a simple multiplier from Table 2. Housing in the West costs 19% more than the national average (multiplier 1.19); housing in the Midwest costs 15% less (multiplier 0.85).

Table 1: Average annual spending and income by age group (U.S., BLS 2024)

| | Under 25 | 25–34 | 35–44 | 45–54 | 55–64 | 65–74 | 75+ |
|---|---|---|---|---|---|---|---|
| Income | 48,514 | 102,494 | 128,285 | 141,121 | 121,571 | 75,460 | 56,028 |
| Food | 7,215 | 9,630 | 12,460 | 12,772 | 10,214 | 8,483 | 7,168 |
| Housing | 16,853 | 26,380 | 30,369 | 30,747 | 27,019 | 22,329 | 21,999 |
| Transport | 9,243 | 12,802 | 15,581 | 17,184 | 15,085 | 11,414 | 6,855 |
| Healthcare | 1,485 | 3,825 | 5,949 | 6,748 | 6,711 | 7,715 | 7,918 |
| Insurance | 4,428 | 10,701 | 13,465 | 14,948 | 12,172 | 4,579 | 1,908 |

Table 2: Regional cost-of-living multipliers (U.S., BLS 2024). Values above 1.0 mean higher than national average.

| Region | Food | Housing | Transport | Healthcare | Insurance |
|---|---|---|---|---|---|
| Northeast | 1.05 | 1.14 | 0.94 | 1.08 | 1.02 |
| Midwest | 0.95 | 0.85 | 0.98 | 0.96 | 1.01 |
| South | 0.97 | 0.92 | 1.03 | 0.98 | 0.99 |
| West | 1.04 | 1.19 | 1.05 | 1.00 | 0.98 |

With regard to the UK profiles, we rely on corresponding ONS survey data, with different regional factors for England and Wales. An important structural note: health spend is essentially zero for most households due to universal coverage provided by the NHS (National Health Service [UK]), funded by taxation rather than insurance.

### 1.4 Estimating the Income Elasticities

Elasticities are the key parameters of our model, and they inform us on the growth of spending with income. They were obtained using household survey data from the U.K., which is the only source that distinguishes between spending broken down by both age groups and income levels (five levels from lowest to highest). While one might be concerned that using U.K. data to calculate elasticities, Engel's law (a law that states the more you make, the smaller percentage of income goes towards necessities) holds true among all countries studied. This is a universal phenomenon and we simply use U.K. because of the data it provides which generally transfers well to other countries.

We used a power-law relationship for spending, which means that spending increases proportionally to income raised to a certain power, rather than a linear relationship. In a linear relationship, getting a $10,000 raise affects your spending on food the same amount whether you make $30,000 or $300,000 per year. But the power-law relationship recognizes that the same raise will have a greater effect on the spending of someone who is low-income than on someone who is high-income. When tested on the U.K. data, the power-law model explained 87–97% of the variation in spending ($R^2$ values in Table 3), compared to below 75% for linear fits.

To calculate these elasticities, we employed a standard statistical technique: log-log regression. By taking the logarithm of spending and income data, any power law relationship will appear as a straight line on a log-log plot, and the slope of this line will give us the elasticity we seek. We controlled for age group in the regression because we know that spending patterns by older adults will increase with age even if their income does not change.

Table 3: How much does spending change when income changes? (Elasticities from U.K. data)

| Category | Elasticity | Fit ($R^2$) | What this means |
|---|---|---|---|
| Food | 0.33 | 0.94 | A 10% income increase leads to only 3.3% more food spending. Confirms Engel's Law: richer households spend a smaller share on food. |
| Housing | 0.16 | 0.87 | Nearly fixed. A raise doesn't change your rent or mortgage. |
| Transport | 0.74 | 0.97 | Responsive. Higher earners buy more expensive cars and commute farther. |
| Healthcare | 0.7 | N/A | Moderate response. Estimated from economics literature since U.K. has public healthcare. |
| Insurance/Pensions | 0.80 | 0.97 | Highly responsive. Retirement contributions scale almost proportionally with income. |

The key insight is that every elasticity (except healthcare) is below 1.0 (Figure 1). These categories are necessities, which means their cost rises at a slower rate than your income. The difference is what is known as disposable income. This is because as your income rises, a smaller percentage goes towards the necessities. We use 0.7 for healthcare as an estimate from economics research. [9]

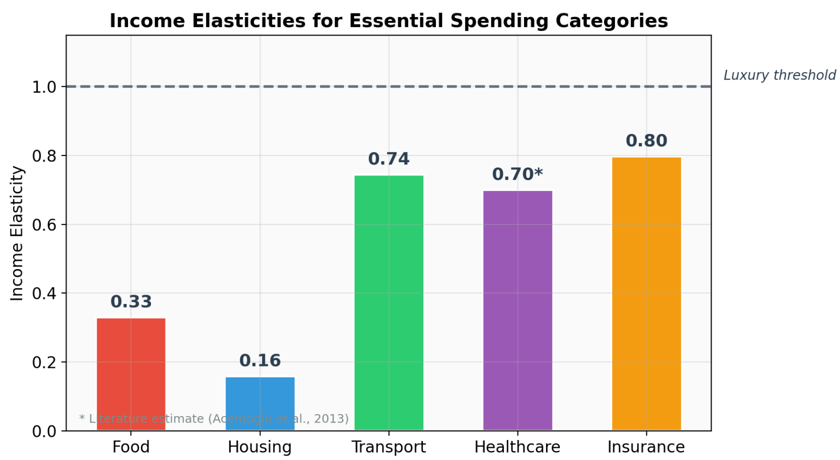


Figure 1: All five elasticities fall below 1.0, confirming that essential spending grows more slowly than income. This is what creates disposable income.

## 1.5 Results

We tested the model on six people with different ages, salaries, regions, and countries. The results (Table 4) reveal large disparities.

Table 4: Disposable income estimates for six profiles (all values in USD)

| Profile | Description | Salary | Total Costs | **Disposable Income** |
|---|---|---|---|---|
| A | Age 27, Northeast, US | $65,000 | $69,559 | **−$4,559** |
| B | Age 42, Midwest, US | $110,000 | $96,569 | **$13,431** |
| C | Age 60, South, US | $85,000 | $75,124 | **$9,876** |
| D | Age 22, West, US | $35,000 | $46,620 | **−$11,620** |
| E | Age 28, England, UK | $48,260 | $32,380 | **$15,880** |
| F | Age 45, Wales, UK | $69,850 | $40,868 | **$28,982** |

Two of the six profiles have negative disposable income, meaning they spend more than they earn after taxes. Profile D is the worst off, with the individual having the lowest disposable income at -$11,620 after earning $35,000 at the age of 22 and living in the West. Three factors contribute to this person's situation: the individual earns the least among the population below the age of 25, averaging $48,514. The region also has the highest housing costs in the country, which are 19% higher than the average. Young adults also tend to live in the most expensive urban areas.

A different picture is given when looking at the U.K. profiles. For the mid-career worker in Wales making $69,850, they still have access to almost $29,000 in disposable income, more than twice what Profile B has access to, despite making $40,000 less than Profile B. It is not a coincidence that this is the case; the cost of housing is significantly lower, and the NHS cancels out the cost of health care altogether.

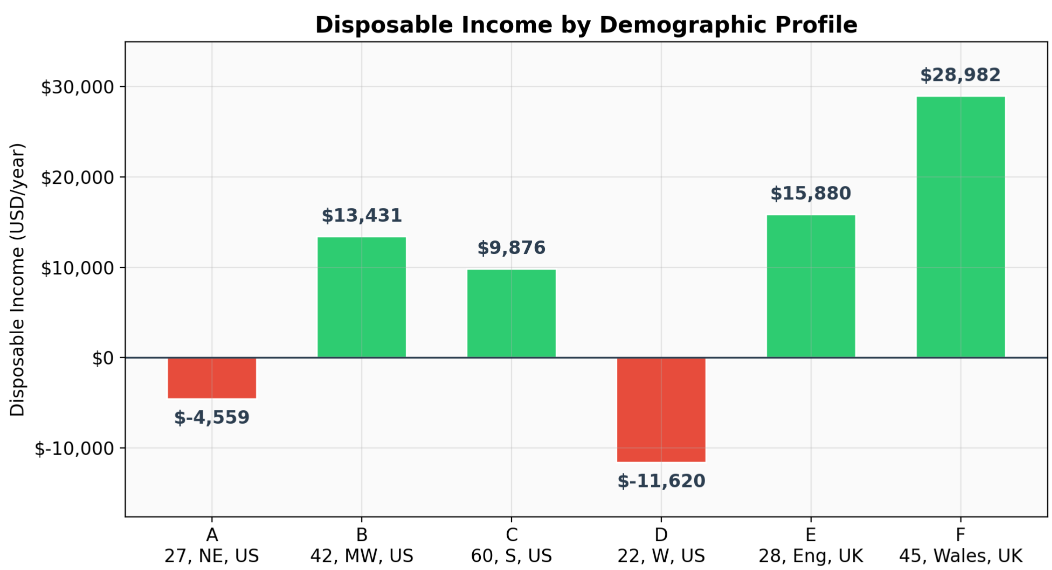


Figure 2: Disposable income varies from negative $11,620 to positive $28,982. Negative values (red) mean essentials exceed after-tax income. U.K. profiles consistently perform better than U.S. profiles at similar salary levels.

Figure 3 breaks down where the money goes. Housing is the biggest expense for all the profiles in the US, accounting for 30-50% of the after-tax income. For the U.K., the total essential expenses are roughly half of the US, mainly due to the lower housing costs and lack of any healthcare expenses.

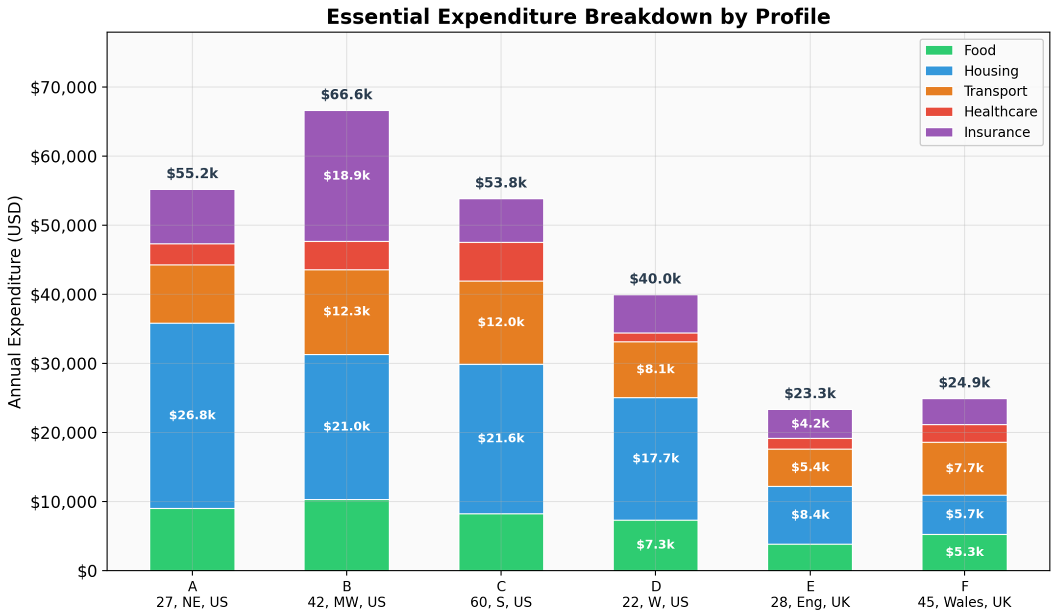


Figure 3: Expenditure breakdown by category. Housing dominates U.S. budgets. U.K. totals are roughly half of U.S. levels.

### 1.6 Sensitivity Analysis

We also tested the sensitivity of our results to changes in our elasticity estimates by moving each of them up or down by 0.10. Housing was the most volatile of the bunch, moving around \$1,200 on Profile A, which isn't surprising given that it's the largest expense category of all. But the bottom line is the same under all scenarios: Profiles A and D remain negative, and the U.S./U.K. difference holds up.

Table 5: Sensitivity of Profile A's disposable income to $\pm 0.10$ elasticity changes

| Category | Elasticity | DI Change ($\pm 0.10$) |
|---|---|---|
| Housing | 0.16 | ±\$1,200 |
| Insurance/Pensions | 0.80 | ±\$800 |
| Transport | 0.74 | ±\$500 |
| Healthcare | 0.70 | ±\$300 |
| Food | 0.33 | ±\$200 |

### 1.7 Strengths and Limitations

Strengths: The study is based on actual government statistics for both countries and follows the power law model that is often used in economic theory with very strong statistical correlation with $R^2$ values above 0.86.

Limitations: The study is based on the assumption that there is only one income earner in the family. The 5% state tax is an approximation. The model does not include individual factors such as choosing to live in a less expensive apartment.

# 2 Q2: Know the Spread

## 2.1 Problem

By using disposable income we found in Q1, we can estimate the net earnings of an individual in a given year from online sports betting. This model relies on two major factors, which are demographics (determining whether a person will gamble) and risk tolerance (determining the extent of the gamble.)

## 2.2 Background: How Sportsbooks Make Money

Before we get to the model, it's worth understanding *why* gamblers lose money on average. Sportsbooks have a profit margin built into every bet, referred to as the house edge. The house edge can vary dramatically depending on the type of bet.

Simple spread bets are the most common type. The sportsbook establishes the point spread so that both sides have an equal probability of winning and then makes money by taking a commission called the "vig." The odds are usually "-110" meaning the person betting risks 110 to win 100. The sportsbook has even odds so they make 110 from the losing bets and pay 100 to the winning ones. They make around 10 for every 210 bet. That is a 4.5% house edge.

A parlay is a bet that involves a collection of picks that are combined into a single wager. In a parlay, all the picks have to win in order for the parlay to win. That is the allure of a parlay: the potential to win a lot. However, the edge is much deeper. If the odds on a 3-item parlay are standard, the actual probability of winning all 3 events is about 10.8%, but the payout is based on the assumption that the odds are about 9%. That means the edge is about 14%. If you go to 5-item parlays, the edge can exceed 25%. Sportsbooks aggressively market parlays because they are much more profitable.

This is the main point we want to get across with our model: not only do riskier gamblers bet larger amounts, but they also make riskier types of bets with much higher house edges. It's a two-fold disadvantage.

## 2.3 Assumptions

1. Wagering is proportional to disposable income. The higher the disposable income of the individual, the higher the amount they would wager. Annual wagering can be represented as a fraction of disposable income and risk tolerance.

2. Risk tolerance controls two behaviors simultaneously: one related to the amount of income risked, and the other related to the style of betting. On a scale of 1 to 5, risk tolerance determines not only the proportion of income risked but also the style of betting. The conservative bettor (Risk 1) seeks simple spread bets with low house edges, while the daring bettor (Risk 5) seeks parlays with larger house edges.

3. Bets within a year are independent. We do not model chasing losses or bankroll management strategies, though we acknowledge these as limitations.

4. Industry calibration. The base fraction used to calculate the wagering losses is set such that the total U.S. losses match the reported $15 billion in 2025 sports betting revenue.

## 2.4 The Model

The expected annual loss for a gambler is:

$$\boxed{\text{Expected Loss} = \text{Annual Wager} \times \text{House Edge}} \tag{2}$$

where the annual wager depends on disposable income and risk level:

$$\text{Annual Wager} = \text{Disposable Income} \times \beta \times \text{Risk Multiplier}$$

The $\beta$ figure represents the amount that the baseline gambler at risk 1 is willing to bet as a percentage of their disposable income. Using the $15 billion figure from assumption 4, we calibrated this number to be 0.10. This increases as the risk increases. The house edge also increases as the risk increases because more risky gamblers tend to favor the types of bets that have the highest built-in margins.

Table 6 shows the full parameter set. Each of these has its basis in the bet type analysis described earlier, with the 4.5% house edge of Risk 1 based directly on the standard -110 odds, the 8% house edge of Risk 3 based on a mix of spread bets and two-leg parlays, etc.

Table 6: How risk tolerance maps to gambling behavior

| Risk Level | Wager Multiplier | House Edge | Typical Bet Type |
|---|---|---|---|
| 1 — Conservative | 1× | 4.5% | Standard spread bets |
| 2 — Cautious | 2× | 5.5% | Spreads plus some prop bets |
| 3 — Moderate | 4× | 8.0% | Mix including small parlays |
| 4 — Aggressive | 7× | 12.0% | Multi-leg parlays |
| 5 — High Risk | 12× | 18.0% | Longshot and exotic parlays |

The wager multipliers follow a kind of geometric progression, with each step roughly doubling the last. Of course, this is similar to the observed reality: aggressive gamblers do not merely increase their wagers somewhat: They increase them substantially and make several wagers daily at much higher stakes.

Special case: Negative Disposable Income. Some gamblers gamble despite their lack of resources, using credit cards to pay bets or dipping into their savings. For all individuals where DI $\leq 0$, a base wager of $2,000 per year is used, amplified by the risk multiplier to account for problem gambling behavior.

How we tuned the model. The initial wagering fraction was set at 10%, and we then validated the degree to which the total predictions correlate to the actual data. By entering the Census population figures and the participation rates given below, the model projects total U.S. gambling losses to be around $15.5 billion—within 3% of the actual industry revenues of $15 billion. That level of correlation to actual data makes us confident that the predictions at the individual level are reasonable.

## 2.5 Who Gambles?

Gambling participation is heavily dependent on age and gender.

Table 7: Gambling participation rates (fraction with active betting accounts)

| Age Group | Men | Women |
|---|---|---|
| 18–24 | 45% | 15% |
| 25–34 | 52% | 18% |
| 35–49 | 48% | 14% |
| 50–64 | 25% | 8% |
| 65+ | 10% | 3% |

This translates into a total of 58 million active American gamblers, which agrees with the stated 22%. The gender gap in gambling participation among the young is worth noting, as males are three times more likely to gamble than females.

## 2.6 Results

We applied the model to six profiles. Table 8 shows how the same disposable income leads to vastly different outcomes depending on risk level.

Table 8: Expected annual gambling losses for six profiles

| Profile | Disp. Income | Risk | Annual Wager | **Expected Loss** |
|---|---|---|---|---|
| Young male, cautious | $13,431 | 1 | $1,343 | **$60** |
| Young male, moderate | $13,431 | 3 | $5,372 | **$430** |
| Young male, high risk | $13,431 | 5 | $16,117 | **$2,901** |
| Mid-career female (UK) | $28,982 | 2 | $5,796 | **$319** |
| Senior, conservative | $9,876 | 1 | $988 | **$44** |
| Young, negative DI | −$11,620 | 4 | $14,000 | **$1,680** |

The first three lines stand out the most. It is the same person with the same earnings, and going from Risk 1 to Risk 5 increases the amount of money that is expected to be lost 48 times, from $60 to $2,901. At Risk 5, this person is shelling out 22% of their entire disposable income. It is dramatic math, and this is because of the compounding nature of the risk. A Risk 5 gambler is betting 12 times as much and faces four times the house edge of the Risk 1 gambler.

Figure 4 shows this scaling visually. The relationship is not linear because losses accelerate as risk increases because both wagering and house edge grow simultaneously.

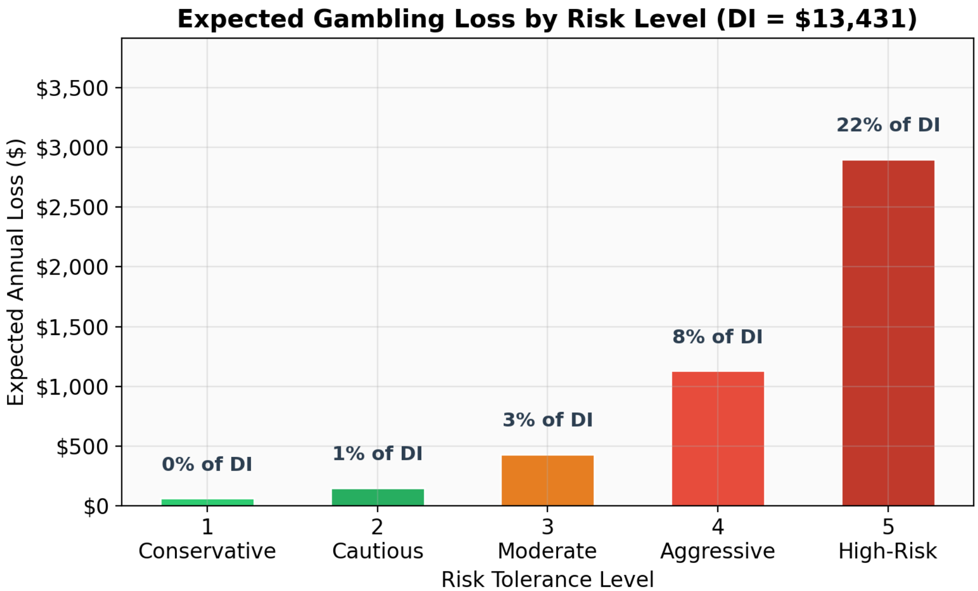


Figure 4: Expected annual loss by risk level for someone with $13,431 in disposable income. The percentage labels show losses as a share of DI: 0.4% at Risk 1, escalating to 21.6% at Risk 5.

Figure 5 shows the demographic picture. The largest expected losses in the whole population (including non-gamblers) are among young men in the 25- to 34-year-old group, totaling $183 per person per year. This is due to their high participation rate (52%), risk-taking behavior, and sufficient discretionary income to support significant wagering amounts. Women and the elderly make minimal contributions to the total losses.

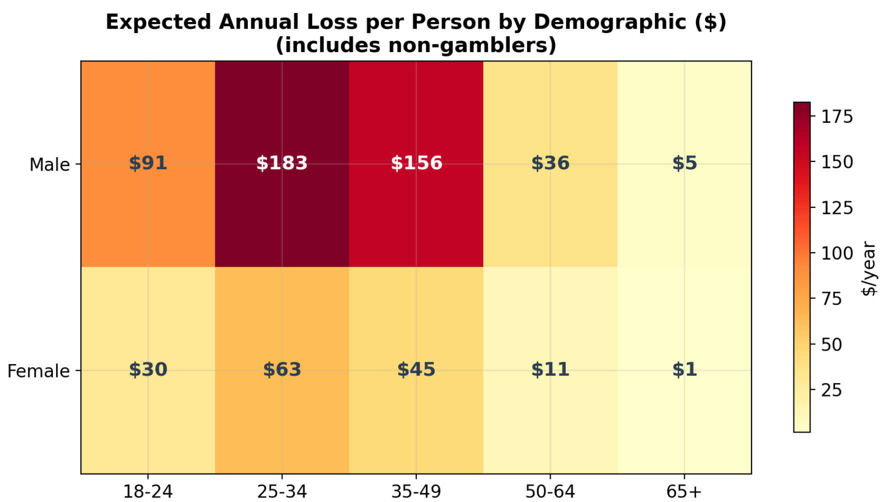


Figure 5: Average annual loss per person by age and gender (including non-gamblers). Darker cells indicate higher losses. Young men 25–34 are the most affected demographic.

## 2.7 Outcome Variability

The expected loss is the average over the long term, but the results can vary significantly from year to year. To account for this, we ran 10,000 simulations for each risk level to model individual years as individual bets with reasonable payout schemes.

Table 9: Range of actual annual outcomes from 10,000 simulated seasons

| Risk Level | Median Outcome | SD | Middle 90% Range | Chance of Profit |
|---|---|---|---|---|
| Risk 1 | −\$60 | \$367 | −\$350 to +\$245 | 31% |
| Risk 3 | −\$393 | \$1,438 | −\$1,480 to +\$460 | 16% |
| Risk 5 | −\$3,023 | \$5,765 | −\$7,670 to +\$1,210 | 4% |

Figure 6 shows the full distribution. Conservative gamblers tend to linger around the break-even point, with a safe 31% probability of ending the year with a profit. On the other hand, the high-risk gamblers face a wide, left-skewed distribution where the majority of the outcomes lie on the left side of the graph, but the occasional winning outcomes provide the excitement associated with risky gambling. The math, however, behind gambling is simple: the house will always win in the long run.

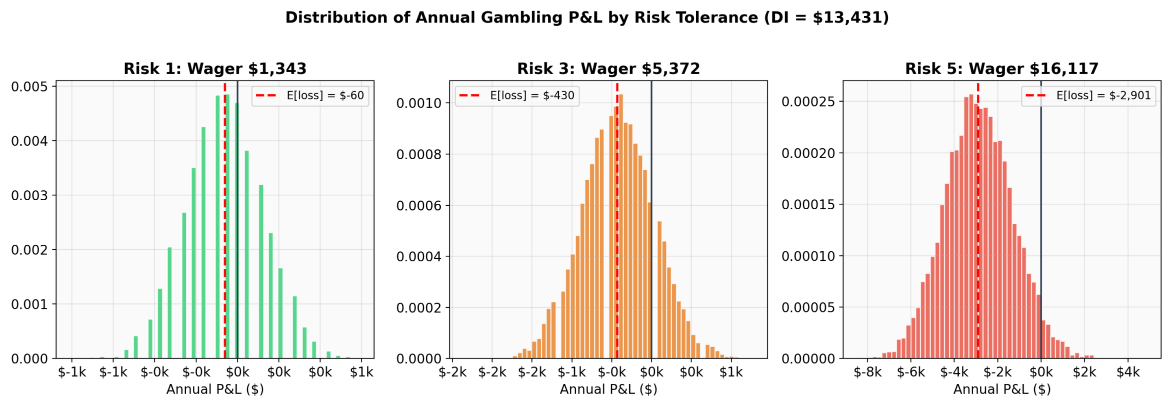


Figure 6: Distribution of annual profit/loss for three risk levels. The vertical dashed lines mark expected losses; the black line marks break-even. Higher risk widens the distribution and shifts it sharply left.

## 2.8 Strengths and Limitations

Strengths: The house edge is based on the math behind the betting prices, not on any assumptions. The model is matched to actual industry revenue within 3%. The model gives complete outcome distributions, not just averages.

Limitations: We assume a fixed risk tolerance, so we do not model the behavior of escalating bets over time. We do not model sportsbooks' promotional offers, such as the "free bets" used to cushion early losses. We do not model the assumption of independent bets, as this would not account for "chasing" behavior.

# 3 Q3: Don't Break the Bank

## 3.1 Problem

We are moving from examining the yearly losses (Q2) to examining the significance of the losses in the long run. Our aim is to express the monetary cost of gambling in a way that the common public can easily understand. To achieve this, we use a stochastic Monte Carlo simulation that examines the savings of households as gambling slowly drains their resources over a period of 30 years, as well as two additional analyses that compare it to how people spend on entertainment and a risk assessment.

## 3.2 Why a Simulation Model?

A simple math problem that anyone can do is: if you lose $430 per year for 30 years, that's a total of $12,900. But that's not the entire problem; the problem is this doesn't address how money actually grows and compounds over time. There's also the fact that the returns on investments or the results of gambling aren't always certain; while the simple math problem would give one definite answer, the results of a simulation would give a range of possible outcomes, including the possibility of going into debt.

## 3.3 Assumptions

1. Baseline savings rate of 20%: Without gambling, the individual would save 20% of their disposable income. This is the standard recommendation given by most financial experts (50/30/20 rule: 50% needs, 30% wants, 20% savings). For Profile B, for example, the individual would have the potential to save approximately $2,686 per year since they have a disposable income of $13,431.
2. Random investment returns: The money saved grows every year by investment returns that vary randomly in a normal distribution with an average of 7% and a standard deviation of 15%. This reflects the historical behavior of the U.S. stock market over the long run, which averages around 7%, though in a given year the investment return could be as high as +25% or as low as -15%.
3. Each year's gambling losses are drawn from the Q2 outcome distribution rather than being based on the average. This is due to the variability that was seen in Q2, where the gambler may make more or less than the average in a given year.
4. Gambling losses have a direct impact on savings. The net savings contribution each year is given by the baseline savings minus the gambling loss of that year. If the gambling loss is larger than the baseline savings, it means that not only is the person failing to save, but they are actually losing wealth and going into debt.
5. Consistent Behavior. The individual maintains their risk tolerance level across thirty years. While we recognize that this is an oversimplification, some gamblers will escalate their activity while others will stop, note the long-term progression of consistent behavior at each risk tolerance level.
6. Starting from zero. All individuals begin with $0 in savings (e.g., starting their financial life at age 25).

### 3.4 The Model

Each year, household net savings update according to:

$$\boxed{\text{Savings}_{t+1} = \text{Savings}_t \times (1 + r_t) + \text{Annual Saving} - \text{Gambling Loss}_t} \quad (3)$$

This equation has three parts:

Growth of existing savings: Whatever has been accumulated so far grows (or shrinks) with the year's investment return. In a good market year ($r_t = +15\%$), $100,000 in savings becomes $115,000. In a bad year ($r_t = -10\%$), it drops to $90,000.

New savings contribution: Each year, the person saves $2,686, which is 20% of their disposable income. This is the amount that the person would save if they did not gamble.

Loss from gambling: we subtract the actual loss of the current year. The usual average loss for a Risk 3 gambler amounts to around $430, though the actual result of the gambler's gamble can vary from a gain of $460 to a loss of $1,480 in a given year, depending on the distribution of the Q2 simulation. If the losses from gambling exceed $2,686, the gambler's failure to save means not only a lack of savings but an expenditure of his wealth as well.

We ran 10,000 simulations per case: no gambling, Risk 1, Risk 3, and Risk 5. Each simulation had its own independent random return and gambling result per year, producing 10,000 unique 30-year financial histories.

We select Profile B as the representative individual, who is 42 years old, makes $110,000 annually, lives in the Midwest, and has $13,431 in disposable income. We conducted 10,000 simulations for each scenario: no gambling, risk 1, risk 3, and risk 5. Each simulation has an independent yearly random return and gambling outcome, providing 10,000 different financial histories for the individual over 30 years.

### 3.5 Results: What Happens to Your Savings?

Figure 7 shows the median savings trajectory (solid line) for each scenario, with the shaded part showing the middle 50% of outcomes. The contrast is significant.

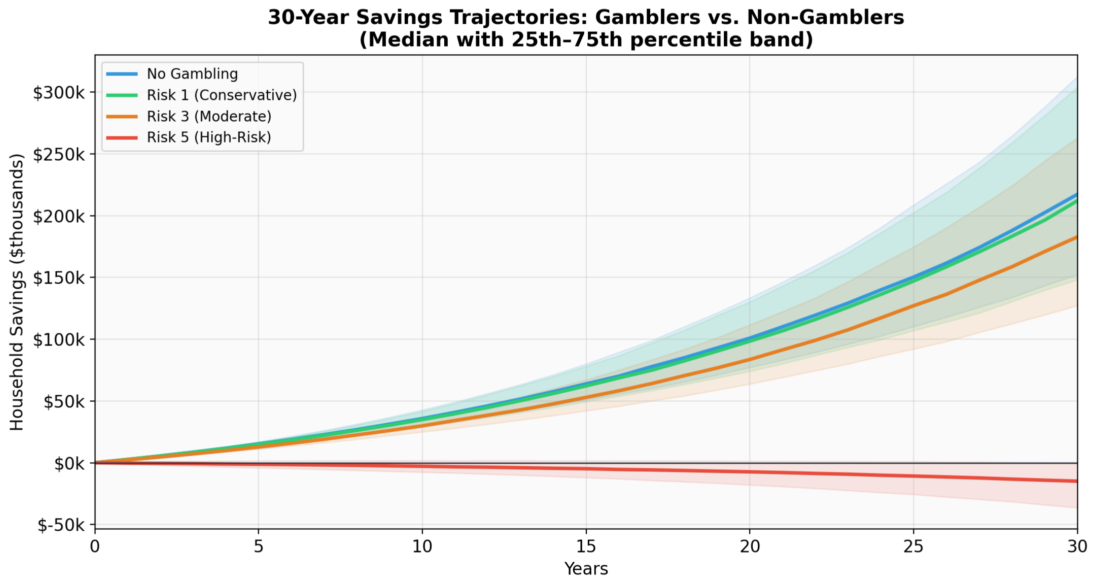


Figure 7: 30-year savings trajectories. Solid lines show medians; shaded bands show the 25th–75th percentile range. The non-gambler and Risk 1 lines nearly overlap. Risk 3 falls behind gradually. Risk 5 sinks below zero.

Table 10: 30-year financial outcomes across 10,000 simulated lifetimes

| Scenario | Median Savings | Worst 10% | Best 10% | Chance of Debt |
|---|---:|---:|---:|---:|
| No gambling | $217,388 | $111,986 | $437,587 | 0.0% |
| Risk 1 (conservative) | $212,198 | $108,194 | $426,156 | 0.0% |
| Risk 3 (moderate) | $182,876 | $94,812 | $370,292 | 0.0% |
| Risk 5 (high-risk) | −$14,922 | −$64,277 | $17,883 | **74.3%** |

Non gambler: If you save $2,686 every year, and your average return is 7%, you'll be able to accumulate a median amount of $217,000 by the time you retire in 30 years. Even during the worst stock markets, the 10th percentile will be $112,000, and you'll never be in debt. Risk 1 (conservative gambler): The median drops by $5,000 relative to the non-gambler. In this category, gambling makes little impact on financial well-being. An expected loss of $60 per year can easily be handled.

Risk 3 (moderate gambler): The gap now grows by $35,000. While it may not seem disastrous, it quietly eats away at 16% of one's savings over a working lifetime. While each year may not seem significant, it's a significant impact nonetheless.

Risk 5 (High Risk Gambler): The expected outcome is a net loss of 15,000 dollars. This means that the majority of the high-risk gamblers not only fail to save but also go into debt. At the end of 30 years, there is a 74.3% probability that the high-risk gamblers will be in net debt. Even the top 10%, who have the best of market conditions and gambling odds, accumulate only up to $18,000, which is less than a tenth of what the non-gambler would have saved. This is because the expected losses from gambling, which is around $2,901 per year, far exceed the initial savings of $2,686 per year.

Figure 8 shows how the probability of being in debt evolves over time. For Risk 5, it crosses 50% within the first two years and climbs to 74% by year 30. All other scenarios remain at 0% throughout.

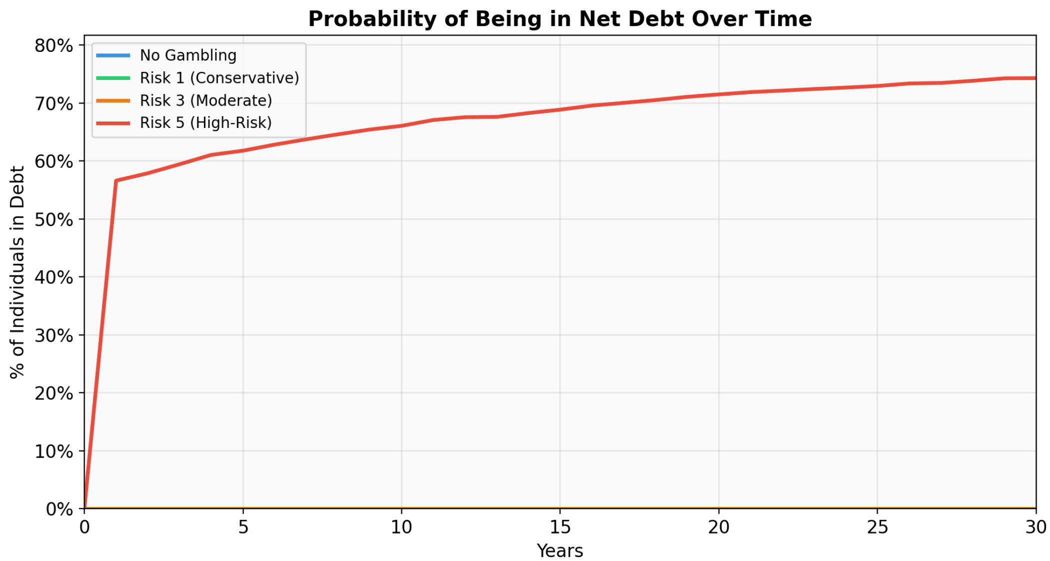


Figure 8: Probability of being in net debt over time. High-risk gamblers cross 50% almost immediately and reach 74% after 30 years. Conservative and moderate gamblers never enter debt territory.

Figure 9 shows the wealth gap created by the difference between the two individuals, the amount of money the non-gambler has more than the gambler at any given time. The longer the time period, the more the gap increases due to the effects of compound interest: gambling 20-30 years is not the same as 1-10 years. The high-risk gambler is 232,000 dollars poorer by the time 30 years has passed.

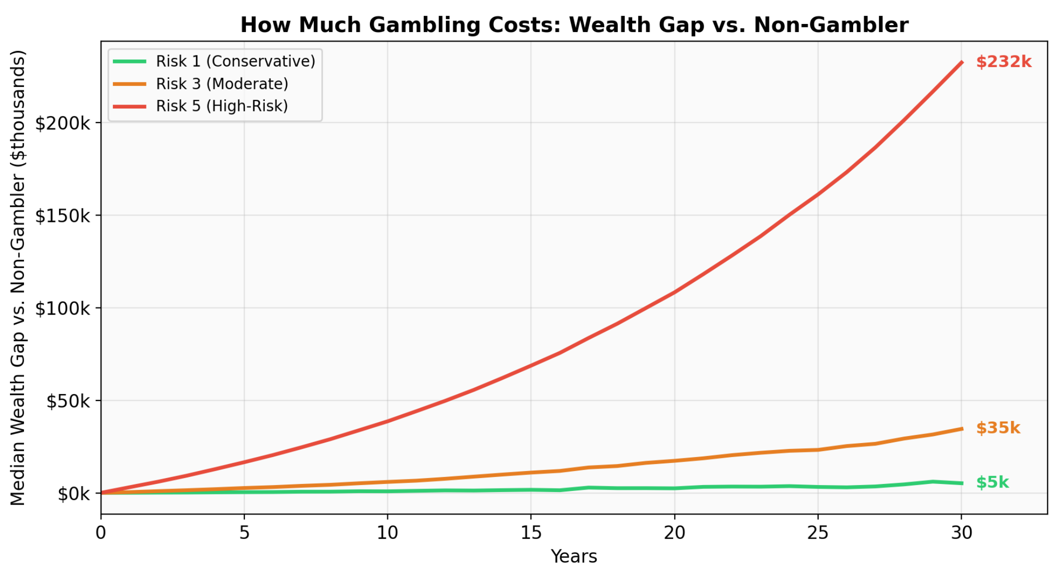


Figure 9: Median wealth gap compared to a non-gambler. The gap accelerates over time due to compound interest. After 30 years, a high-risk gambler has $232,000 less than a non-gambler.

This is particularly concerning because the segment of gamblers most vulnerable to heavy gambling habits. Namely, young men between 25 and 34 years of age (Q2)also happens to have the longest time ahead of them. A 25-year-old gambler with a 30-year heavy gambling habit will have $200,000 in debt by the time he reaches 55 years of age, while a non-gambler will have saved more than $200,000 by that time.

### 3.6 Supplementary Analysis: Gambling vs. Entertainment Spending

To provide additional context for the general public, we compared annual gambling losses to other common entertainment expenses (Figure 10).

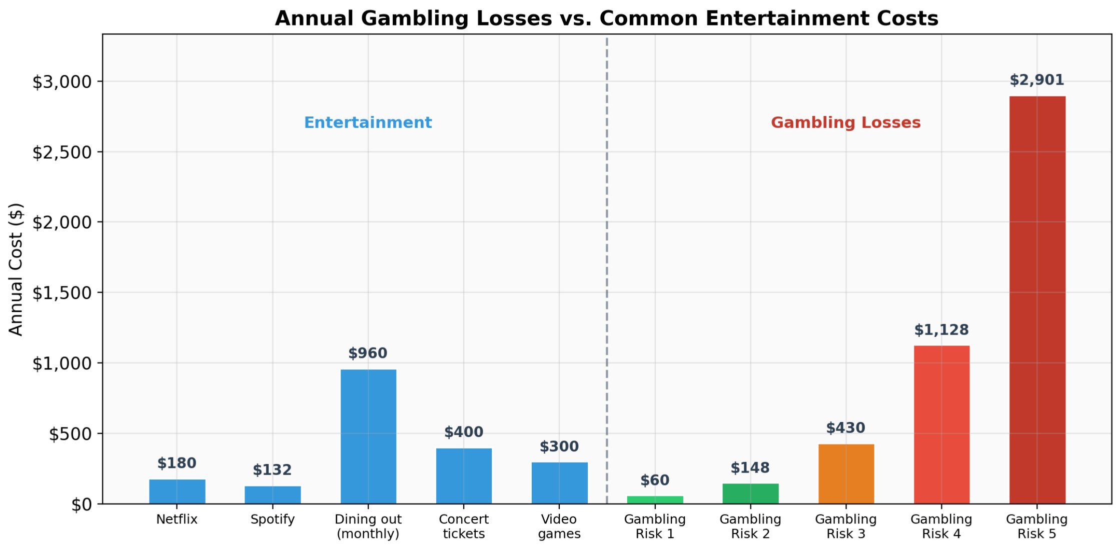


Figure 10: Annual gambling losses compared to common entertainment costs. At Risk 1–2, gambling is within a typical entertainment budget. At Risk 5, it exceeds all listed entertainment combined.

For Risk 1, the cost of gambling is only around $60 per year, which is way less than the annual subscription fee of $132 to Spotify. When the risk level goes up to Risk 3, the $430 spent on gambling looks more like the amount spent on concert tickets or video games per year. At Risk 5, the $2,901 spent on gambling is way more than the total amount spent on video games, concerts, Netflix, Spotify, and dining out per year, which amounts to a total of $1,972 spent on all these forms of entertainment combined. At this risk level, gambling is no longer a source of entertainment but the biggest expense on the household budget.

### 3.7 Supplementary Analysis: How Many People Are Getting Hurt?

Using Q2's simulations (50,000 trials per risk level), we estimated what fraction of gamblers exceed various financial harm thresholds in a single year.

Table 11: Fraction of gamblers who lose more than a given share of their disposable income in one year

| | Lose >5% | Lose >10% | Lose >25% | Lose >50% |
|---|---|---|---|---|
| Risk 1 | <1% | <1% | 0% | 0% |
| Risk 3 | 33% | 2% | <1% | 0% |
| Risk 5 | 92% | 84% | 40% | <1% |

For the conservative gamblers, almost nobody crosses any harm threshold. However, for Risk 5 gamblers, the situation is quite the reverse: a massive 84% lose more than 10% of their disposable income, and 40% lose more than a quarter. Losing a quarter of $13,431 is not a small amount; it is equivalent to losing $3,358, or two months' rent for a typical US citizen. The increase from Risk 3 to Risk 5 is not a gradual increase; it is a massive increase of 42 times for those who lose more than 10%, from 2% to 84%.

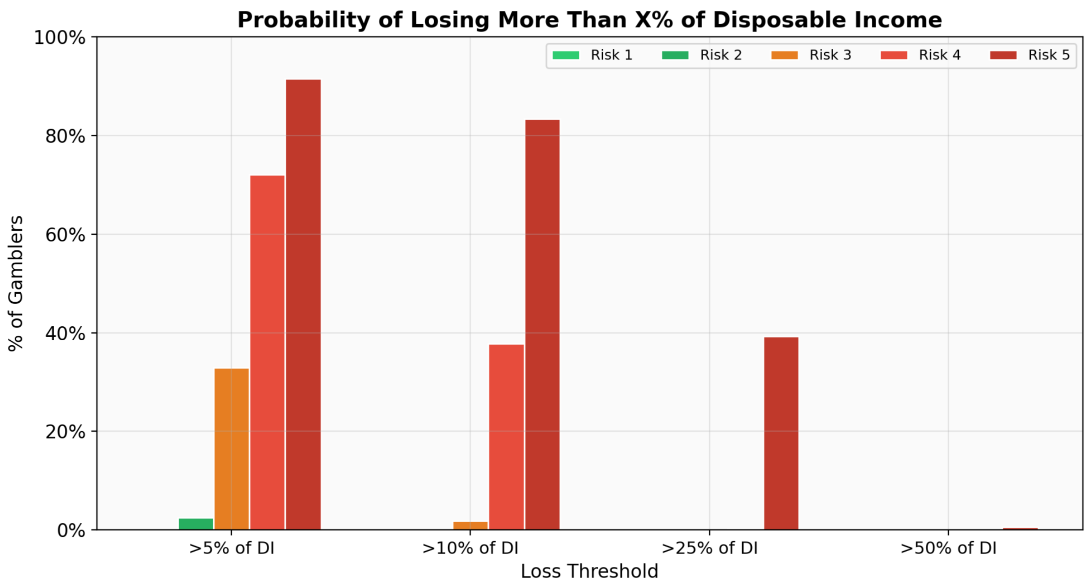


Figure 11: Probability of exceeding each loss threshold by risk level. The transition from moderate to aggressive gambling represents a sharp discontinuity in financial danger.

### 3.8 Synthesis: Should Society Be Concerned?

The long-term wealth trajectories, entertainment comparisons, and population risk analyses all tell a consistent story.

Table 12: Summary of findings across all three analyses

| Risk Level | Compared to Entertainment | 30-Year Wealth Impact | Single-Year Harm |
|---|---|---|---|
| 1–2 (Low) | Within budget; cheaper than most subscriptions | Negligible: $5K gap (2%) | <1% exceed any threshold |
| 3 (Moderate) | One of the larger line items (like concerts) | Meaningful: $35K gap (16%) | 33% lose >5% of DI |
| 4–5 (High) | More than all other entertainment combined | Devastating: **median in debt** (74% chance) | 84% lose >10% of DI |

Low-risk gamblers are not a financial concern. They fit comfortably into a normal entertainment budget and have a negligible effect on savings. There is no strong mathematical rationale to justify society's concern about this group.

Moderate gambling results in actual but manageable costs. Over 30 years, the resulting wealth decrease of $35,000 is significant and graspable, though none of these people will be ruined. The proper course of action is education and awareness.

High-risk gamblers are those whose numbers are really painful. This group of people loses more money each year than all other forms of entertainment combined. They have a 75% chance of ending their 30-year working career with net debt and continue to exceed financial harm thresholds every single year. This group of people mostly comprises young men between 18 and 34 years of age, who have the most to lose if compound growth is forfeited.

The key takeaway is that the negative effects of gambling are not spread out across the population; they are concentrated. On average, a gambler loses about $266 per year, which is less than the cost of a month-long Netflix subscription. However, the problem is that the average masks the disproportionate effect that a small percentage of high-risk gamblers have on the $15 billion in revenue that the gambling industry generates, as they consistently ruin their own financial futures. Good policy does not try to stop gambling in its entirety, but instead targets the vulnerable minority that causes the problem in the first place, using income limits on how much they can spend, notifications when they start to go all in on high-edge betting options, and targeting young men to ensure that their recreational activity does not lead to a financial crisis.

### 3.9 Strengths and Limitations

Strengths: In Q3, we use the new model stochastic savings simulation instead of simply repeating the results from Q1-Q2. This model accounts for the uncertainty due to both the investment markets and the gambling at the same time. It also provides results instead of a simple point estimate due to the use of 10,000 trajectories.

Limitations: Our steady risk tolerance over 30 years may not reflect real gamblers who may escalate or fold their bets. Our 20% base savings rate may be more of an aspirational value than a current reality, personal savings rates in the US are currently closer to 4-5%, so our non-gambling baseline may actually be overly optimistic. Ironically, this makes our estimates of the impact of gambling somewhat conservative because of the lower savings cushion to fall back on in the face of losses. We do not capture how people might respond by decreasing their gambling as their wealth declines.

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